SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	**ITA0512**	**Course Title:**	**Computer Vision**
CO Assessed:	**CO4 – Articulation (combining methods for evaluation) P4**	**Assessment:**	**Assessment Tool 2 – Industry Problem Solving Task**
Weightage:	**20% of CIA**	**Total Marks:**	**100**
CO4 Statement:	*Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL5)*		
Student Name:	S. Susannal Devapriyam	**Reg. No.:**	*192525398*
Section / Batch:	B. Tech AI & ML	**Date:**	19 - 08 - 2026

Code of Conduct

I , S. Susannal Devapriyam (192525398) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *S. Susannal Devapriyam*

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**	**Marks**
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context	
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts	
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution	
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools	
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis	
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation	

Question:

9. Mobile Face Detection Trade-off

Constraints:

● Real-time operation required

● Accuracy ≥85%

Options:

Viola-Jones: 80%, fast

Deep Learning: 95%, slow

Compare and evaluate which approach should be chosen. Justify using constraint satisfaction logic**.**

Case Study: Mobile Face Detection Trade-off

Problem Statement:

A mobile face-detection system must operate under two important industry constraints:

- Real-time operation is required.
- Accuracy must be at least 85%.

Two approaches are available:

Approach	**Accuracy**	**Speed**
Viola–Jones	80%	Fast
Deep Learning	95%	Slow

The objective is to compare both approaches and select the one that satisfies the given constraints using **constraint satisfaction logic**.

1. Understanding of the Industry Problem

1.1 Problem Interpretation

Face detection is widely used in mobile applications such as camera applications, authentication systems, augmented reality, security applications, and human-computer interaction.

A mobile face-detection system has limited computational resources compared with a powerful desktop or cloud server. Therefore, the selected algorithm must provide an appropriate balance between accuracy and processing speed**.**

In this case, two constraints are explicitly specified:

1. The system must operate in real time**.**
2. The system must achieve at least 85% accuracy**.**

Therefore, an algorithm cannot be selected merely because it is fast or accurate. It must satisfy the required constraints.

1.2 Understanding the Given Options

Viola–Jones

Viola–Jones is a classical object-detection approach that uses Haar-like features, an integral image, AdaBoost, and a cascade classifier.

Its given performance is:

Accuracy = 80%

Speed = Fast

Therefore, it performs well in terms of speed but does not reach the required accuracy.

Deep Learning

Deep-learning-based face detection uses trained neural networks to learn complex visual patterns.

Its given performance is:

Accuracy = 95%

Speed = Slow

Therefore, it satisfies the required accuracy but may not satisfy the real-time requirement on a constrained mobile device.

1.3 Industry Requirements

The system requirements can be represented as:

Requirement 1:
Real-time operation = Required

Requirement 2:
Accuracy ≥ 85%

Therefore, an acceptable solution must satisfy:

Accuracy ≥ 85%
AND
Real-time operation = YES

This is important because both conditions are mandatory**.**

2. Application of Engineering Concepts

2.1 Constraint Satisfaction Logic

This problem can be treated as a **constraint satisfaction problem**.

Let:

- **A** = Accuracy requirement
- **S** = Speed/real-time requirement

The constraints are:

$A \geq 85\%$

and

$S = \text{Real-time}$

The algorithm is acceptable only when:

$A \geq 85\% \text{AND} S = \text{Real-time}$

2.2 Evaluation of Viola–Jones

Given:

$\text{Accuracy} = 80\%$

Required:

$\text{Accuracy} \geq 85\%$

Comparison:

80<85

Therefore:

Accuracy constraint → FAIL

However:

Speed constraint → PASS

Hence:

Viola–Jones

Accuracy Constraint → FAIL
Real-Time Constraint → PASS

Overall → NOT SATISFIED

Even though Viola–Jones is fast, its 80% accuracy is below the required 85%.

2.3 Evaluation of Deep Learning

Given:

Accuracy=95%

Required:

Accuracy≥85%

Comparison:

95>85

Therefore:

Accuracy constraint → PASS

However:

Speed constraint → FAIL, because the problem states that the deep-learning approach is slow.

Therefore:

Deep Learning

Accuracy Constraint → PASS

Real-Time Constraint → FAIL

Overall → NOT SATISFIED

2.4 Constraint Satisfaction Table

Constraint	Required	Viola–Jones	Deep Learning
Accuracy	≥ 85%	80%	95%
Real-time operation	Required	✅Fast	Slow
Accuracy satisfied?	Yes	No	Yes
Speed satisfied?	Yes	Yes	No
Overall feasibility	Both required	--	--

The table clearly shows that neither given option completely satisfies both constraints.

3. Solution Design

3.1 Why Choosing Directly Is Not Appropriate

At first glance, Deep Learning appears better because it provides 95% accuracy, while Viola–Jones provides only 80%.

However, selecting Deep Learning directly would ignore the real-time requirement.

Similarly, selecting Viola–Jones because it is fast would ignore the minimum accuracy requirement.

Therefore, neither option should be selected unchanged.

4. Proposed Engineering Solution

A practical engineering solution is to use a lightweight deep-learning face detector optimized for mobile hardware**.**

Instead of using a computationally heavy deep-learning model, the system can use a lightweight architecture designed specifically for mobile or edge devices.

The objective is:

High Accuracy
+
Fast Inference
↓
Real-Time Mobile Face Detection

4.1 Proposed Hybrid Optimization

A suitable approach is:

Input Camera Frame
↓
Lightweight Preprocessing
↓
Lightweight Deep-Learning Detector
↓
Face Detection
↓
Confidence Filtering
↓
Real-Time Output

This approach attempts to retain the accuracy advantage of deep learning while reducing its computational cost.

4.2 Model Optimization Techniques

Several engineering techniques can improve deep-learning inference speed:

1. Model Compression

Reducing unnecessary model parameters decreases memory and computational requirements.

2. Quantization

Model weights and activations can use lower numerical precision, reducing computation and memory usage.

3. Pruning

Less important network connections can be removed to make the model more efficient.

4. Hardware Acceleration

Mobile GPUs, NPUs, or other supported accelerators can be used to improve inference speed.

5. Smaller Input Resolution

Processing a suitably reduced image resolution decreases the number of computations required per frame.

5. Alternative Solution: Two-Stage Detection

Another possible design is to use a **two-stage approach**.

Camera Frame
↓
Fast Initial Detection
↓
Candidate Face Regions
↓
Lightweight Accurate Detector
↓
Final Face Detection

The fast detector can quickly identify possible face regions, while a more accurate model processes only those regions.

This can reduce the amount of computation compared with applying a large deep-learning model to the entire image.

6. Why Not Simply Choose Viola–Jones?

Viola–Jones has an important advantage: speed**.**

However:

$80\% < 85\%$

Therefore, it directly violates the minimum accuracy constraint.

For an application where incorrect face detection can affect user experience or security, an accuracy of 80% is insufficient according to the stated requirement.

Thus, Viola–Jones cannot be accepted as the final solution without improvement.

7. Why Not Simply Choose the Given Deep-Learning Model?

The deep-learning approach has:

95% accuracy

which satisfies:

$95\% \geq 85\%$

However, the problem explicitly states that it is **slow**.

If the system must operate in real time, slow inference can cause:

- Delayed detection
- Low frame rate
- Poor user experience
- Increased computational load
- Higher energy consumption

Therefore, the given deep-learning implementation also does not completely satisfy the constraints.

8. Use of Tools

Modern tools can be used to develop and evaluate the mobile face-detection system.

8.1 OpenCV

OpenCV can be used for:

- Image acquisition
- Video-frame processing
- Face detection
- Image preprocessing
- Performance measurement

8.2 Mobile Deep-Learning Frameworks

A mobile-compatible inference framework can deploy optimized neural networks on smartphones.

The model can be converted into an optimized format and executed using available mobile hardware acceleration.

8.3 Hardware Acceleration

Mobile hardware such as:

- GPU
- NPU
- Neural accelerator

can improve inference speed compared with CPU-only processing.

8.4 Performance Measurement

The system should measure:

Accuracy

Accuracy=Total PredictionsCorrect Predictions×100

Processing Speed

Processing speed can be represented using:

FPS=Processing TimeNumber of Frames

A higher FPS indicates faster processing.

9. Analysis and Performance Evaluation

The given values can be analyzed using a simple constraint matrix.

Approach	Accuracy	Speed	Accuracy Constraint	Real-Time Constraint	Decision
Viola–Jones	80%	Fast	No	Yes	Reject
Deep Learning	95%	Slow	Yes	No	Reject as given
Optimized Deep Learning	Target ≥85%	Target Real-Time	Yes	Yes	Preferred

The first two options demonstrate an important engineering trade-off.

Viola–Jones provides speed but insufficient accuracy.

Deep Learning provides accuracy but insufficient speed.

Therefore, the engineering objective should be to optimize the deep-learning approach for mobile deployment**.**

10. Constraint Satisfaction Decision

The decision can be represented as follows:

```
                START
                  |
                  ▼
         Check Accuracy ≥ 85%
               /       \
             NO         YES
             |           |
             ▼           ▼
          Reject        Check
                      Real-Time
                        /   \
                      NO     YES
                      |       |
                      ▼       ▼
                  Optimize   Accept
                             Model
```

Applying this logic:

Viola–Jones

80% Accuracy
↓
Below 85%
↓
REJECT

Deep Learning

95% Accuracy
↓
Above 85%
↓
But Slow
↓
Optimize for Real-Time

11. Recommended Final Solution

The recommended approach is an optimized lightweight deep-learning model deployed with mobile hardware acceleration**.**

The reason is that the deep-learning approach already achieves:

95% accuracy

which is comfortably above the required:

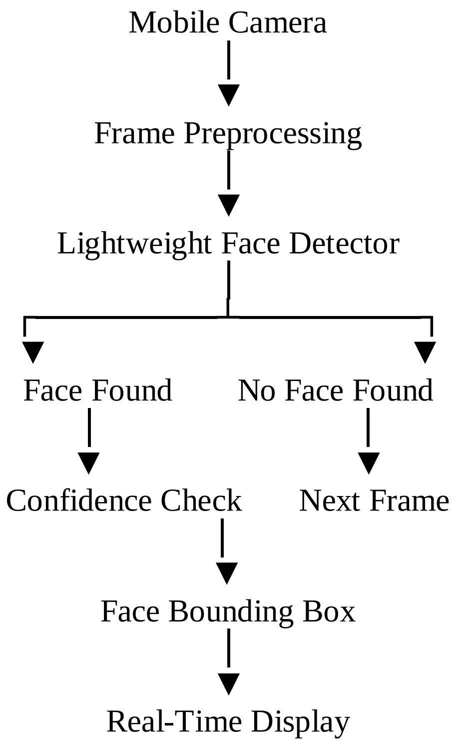
85%

The main engineering challenge is therefore speed optimization.

Techniques such as quantization, pruning, model compression, smaller input resolution, and hardware acceleration can be used to reduce inference time.

12. Practical Mobile Architecture

A practical system can be designed as:



This architecture separates image acquisition, detection, validation, and output.

13. Industry Applications

The optimized approach can be used in:

Mobile Authentication

Face detection can identify the user's face before authentication.

Camera Applications

The system can detect faces for automatic focusing and image enhancement.

Augmented Reality

Real-time face detection can identify facial regions for AR effects.

Security Applications

Face detection can assist in monitoring and access-control systems.

Video Communication

The detector can identify faces during video calls and support intelligent camera features.

14. Advantages of the Proposed Solution

1. Achieves the required accuracy target of at least 85%.
2. Reduces the computational cost of conventional large deep-learning models.
3. Supports real-time mobile processing after suitable optimization.
4. Can utilize mobile GPU or neural acceleration hardware.
5. Provides better accuracy than the given Viola–Jones approach.
6. Offers a practical balance between speed and accuracy.
7. Can be adapted for different mobile applications.

15. Limitations

Although the optimized deep-learning approach is preferred, some limitations remain.

- Model optimization may slightly reduce accuracy.
- Hardware capabilities differ between mobile devices.
- Very small faces may be difficult to detect.
- Poor lighting can reduce detection performance.
- Model conversion may require additional development effort.
- Battery consumption must be considered during continuous detection.

Therefore, the final system should be tested on the actual target mobile device rather than relying only on desktop performance.

16. Final Evaluation

The given approaches can be summarized as:

Viola–Jones

Strength**:** Fast

Weakness**:** 80% accuracy

Constraint Status**:** Fails accuracy requirement

Deep Learning

Strength: 95% accuracy

Weakness: Slow

Constraint Status: Fails real-time requirement as given

Optimized Deep Learning

Strength: Potential to retain high accuracy while improving inference speed

Constraint Status**:** Target solution if measured performance satisfies both constraints

17. Conclusion

The case demonstrates an important accuracy-versus-speed trade-off in mobile Computer Vision.

Viola–Jones is fast enough for real-time operation but provides only 80% accuracy, which is below the required 85%. Therefore, it cannot satisfy the accuracy constraint.

The given Deep Learning approach achieves 95% accuracy, satisfying the accuracy requirement, but its slow processing violates the real-time constraint.

The most appropriate engineering solution is to use a lightweight and optimized deep-learning face detector, combined with techniques such as quantization, pruning, model compression, reduced input resolution, and mobile hardware acceleration.Thus, the decision is based not on choosing the algorithm with the highest accuracy or fastest speed individually, but on satisfying all mandatory system constraints simultaneously**.**